

Model Fit on Historical Data

Examining first vs. second order markov model and interactnig yprev with time or not.

Setup

```
library(here)
library(tidyverse)
library(data.table)
library(arrow)
library(rms)
library(rmsb)
library(furrr)
library(VGAM)
library(Hmisc)
library(ordinal)
library(marginaleffects)
```

Load data

```
df <- read_parquet(here("data", "hosp_long_weeks.parquet")) |>
  mutate(
    tx = factor(tx),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = FALSE)
  ) |>
  select(pat_id, tx, y, yprev, week, ecog_fstcnt, diagnosis, age, gender, albumin,
c_reactive_protein)

df_ypprev <- read_parquet(here("data", "hosp_long_weeks_two_lags.parquet")) |>
  mutate(
    tx = factor(tx),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = FALSE),
    y = ordered(y)
  ) |>
  select(pat_id, tx, y, yprev, ypprev, week, ecog_fstcnt, diagnosis, age, gender,
albumin, c_reactive_protein)

df_daily <- read_parquet(here("data", "hosp_long_daily.parquet")) |>
  mutate(
    tx = factor(tx),
    ecog_fstcnt = factor(ecog_fstcnt, ordered = FALSE),
    y = case_when(
      y == 4 ~ 3,
      y == 5 ~ 4,
      TRUE ~ y),
    yprev = case_when(
      yprev == "4" ~ "3",
      TRUE ~ yprev),
    ypprev = case_when(
      ypprev == "4" ~ "3",
      TRUE ~ ypprev)
```

```

) |>
  select(pat_id, tx, y, yprev, ypprev, day, ecog_fstcnt, diagnosis, age, gender,
         albumin, c_reactive_protein)

```

Weekly States

VGAM constrained

```

d <- as.data.table(df_ypprev)
week_spl_m <- rcs(d$week, 4)

d$week_linear <- week_spl_m[, 1]
d$week_nonlinear_1 <- week_spl_m[, 2]
d$week_nonlinear_2 <- week_spl_m[, 3]

M <- 4

cons <- list(
  # Intercepts of course have separate parameters
  "(Intercept)" = diag(M),

  # Proportional odds terms now get "column of ones" matrix
  "tx" = cbind(P0_effect = 1),
  "yprev" = cbind(P0_effect = 1),
  "week_nonlinear_1" = cbind(P0_effect = 1),
  "week_nonlinear_2" = cbind(P0_effect = 1),

  # The linear component of the spline gets the special constraint
  "week_linear" = cbind(
    baseline_effect = 1,
    linear_trend = 0:(M - 1)
  )
)

# P0 model, first order
-----
fit_po <- vglm(
  y ~ tx + yprev + week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = TRUE, reverse = TRUE),
  data = d,
  constraints = NULL
)

# pP0 for linear time, first order
-----
fit_non_po <- vglm(
  y ~ tx + yprev + week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = FALSE ~ week_linear, reverse = TRUE),
  data = d,
  constraints = cons
)

```

```
Warning in vglm.fitter(x = x, y = y, w = w, offset = offset, Xm2 = Xm2, :  
convergence not obtained in 30 IRLS iterations
```

```
# Doesn't converge well even with constraints  
AIC(fit_po); AIC(fit_non_po)
```

```
[1] 5049.148
```

```
[1] 5046.515
```

```
# --- non-PO with interaction of yprev and linear time ---  
# constraints when we add interaction  
cons_lin_int <- list(  
  # Intercepts of course have separate parameters  
  "(Intercept)"      = diag(M),  
  
  # Proportional odds terms now get "column of ones" matrix  
  "tx"                = cbind(P0_effect = 1),  
  "yprev"             = cbind(P0_effect = 1),  
  "week_nonlinear_1"  = cbind(P0_effect = 1),  
  "week_nonlinear_2"  = cbind(P0_effect = 1),  
  "yprev:week_linear" = cbind(P0_effect = 1),  
  
  # The linear component of the spline gets the special constraint  
  "week_linear" = cbind(  
    baseline_effect = 1,  
    linear_trend    = 0:(M - 1)  
  )  
)  
  
fit_non_po_lin_int <- vglm(  
  y ~ tx + yprev * week_linear + week_nonlinear_1 + week_nonlinear_2,  
  family = cumulative(parallel = FALSE ~ week_linear, reverse = TRUE),  
  data = d,  
  constraints = cons_lin_int  
)  
  
fit_po_lin_int <- vglm(  
  y ~ tx + yprev * week_linear + week_nonlinear_1 + week_nonlinear_2,  
  family = cumulative(parallel = TRUE, reverse = TRUE),  
  data = d  
)  
  
# interaction of yprev with time improves fit by a lot  
# non proportional effect of time does not --> we can likely drop this  
AIC(fit_non_po); AIC(fit_po_lin_int); AIC(fit_non_po_lin_int)
```

```
[1] 5046.515
```

```
[1] 5037.056
```

```
[1] 5036.458
```

```
# --- Second order, full P0 vs partial P0 ---

# constraints when we add interaction
cons_second <- list(
  # Intercepts of course have separate parameters
  "(Intercept)"      = diag(M),

  # Proportional odds terms now get "column of ones" matrix
  "tx"                = cbind(P0_effect = 1),
  "yprev"             = cbind(P0_effect = 1),
  "ypprev"            = cbind(P0_effect = 1),
  "week_nonlinear_1"  = cbind(P0_effect = 1),
  "week_nonlinear_2"  = cbind(P0_effect = 1),

  # The linear component of the spline gets the special constraint
  "week_linear" = cbind(
    baseline_effect = 1,
    linear_trend    = 0:(M - 1)
  )
)

# Doesn't converge well even with constraints
fit_second_non_po <- vglm(
  y ~ tx + yprev + ypprev + week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = FALSE ~ week_linear, reverse = TRUE),
  data = d,
  constraints = cons_second
)
```

```
Warning in vglm.fitter(x = x, y = y, w = w, offset = offset, Xm2 = Xm2, :
convergence not obtained in 30 IRLS iterations
```

```
fit_second_po <- vglm(
  y ~ tx + yprev + ypprev + week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = TRUE, reverse = TRUE),
  data = d
)

# second order fit is substantially better
AIC(fit_po); AIC(fit_second_po)
```

```
[1] 5049.148
```

```
[1] 5042.778
```

```
# second order with interaction of yprev and linear time
-----
cons_second_lin_int <- list(
  # Intercepts of course have separate parameters
  "(Intercept)" = diag(M),

  # Proportional odds terms now get "column of ones" matrix
  "tx" = cbind(P0_effect = 1),
  "yprev" = cbind(P0_effect = 1),
  "ypprev" = cbind(P0_effect = 1),
  "week_nonlinear_1" = cbind(P0_effect = 1),
  "week_nonlinear_2" = cbind(P0_effect = 1),
  "yprev:week_linear" = cbind(P0_effect = 1),
  "ypprev:week_linear" = cbind(P0_effect = 1),

  # The linear component of the spline gets the special constraint
  "week_linear" = cbind(
    baseline_effect = 1,
    linear_trend = 0:(M - 1)
  )
)

fit_second_non_po_lin_int <- vglm(
  y ~ tx + (yprev + ypprev) * week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = FALSE ~ week_linear, reverse = TRUE),
  data = d,
  constraints = cons_second_lin_int
)

fit_second_po_lin_int <- vglm(
  y ~ tx + (yprev + ypprev) * week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = TRUE, reverse = TRUE),
  data = d
)

# strong improvement with interaction of yprev and ypprev with time
# non-proportional effect of time still doesn't improve fit much
AIC(fit_second_po); AIC(fit_second_po_lin_int); AIC(fit_second_non_po_lin_int)
```

```
[1] 5042.778
```

```
[1] 5028.802
```

```
[1] 5028.059
```

```
# second order model, no interaction with time, but interaction between yprev and
yprev -----
cons_second_yprev_int <- list(
  # Intercepts of course have separate parameters
  "(Intercept)" = diag(M),

  # Proportional odds terms now get "column of ones" matrix
```

```

"tx"          = cbind(P0_effect = 1),
"yprev"       = cbind(P0_effect = 1),
"ypprev"      = cbind(P0_effect = 1),
"yprev:ypprev" = cbind(P0_effect = 1),
"week_nonlinear_1" = cbind(P0_effect = 1),
"week_nonlinear_2" = cbind(P0_effect = 1),

# The linear component of the spline gets the special constraint
"week_linear" = cbind(
  baseline_effect = 1,
  linear_trend    = 0:(M - 1)
)
)

fit_second_non_po_yprev_int <- vglm(
  y ~ tx + yprev * ypprev + week_linear + week_nonlinear_1 + week_nonlinear_2,
  family = cumulative(parallel = FALSE ~ week_linear, reverse = TRUE),
  data = d,
  constraints = cons_second_yprev_int
)

```

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Results in intersecting linear predictors

Second order fit seems to be better than first order fit, interaction of y_{prev} and y_{pprev} with previous time improves fit by a lot. Non-linear modelling of time doesn't improve fit much. -> stick to full PO model with interaction of y_{prev} and y_{pprev} with linear time component.